Ground station networks

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1 Ground segment development for CubeSat missions

The rise of innovative, more complex CubeSat missions calls for new solutions to be implemented in the architecture of the ground segment. The exigency of maintaining the costs low is pushing the evolution of ground architectures toward cloud-based systems, in which the ground stations can be used in a flexible way by multiple users just like a network of sensors on the ground. This leads to the identification of a set of requirements for the operation of different missions, which defines standardized communication protocols. For CubeSats, this idea started within the academic world and is now moving toward the creation of commercial networks operated by private entities.

1.1 CubeSat mission overview

CubeSat missions require different types of communications support services from the ground, depending on the number of satellites to be deployed in orbit. These missions can be classified into the following categories, depending on what requirements they imposed on their ground systems:

(a) **Single satellite missions** for either educational projects (mostly universities) or for scientific and/or technology demonstration purposes (research institutions). Examples of this particular type of missions are the ones led by the PolySat laboratory in CalPoly—like CP8—or others deployed through initiatives like NASA's ELANA program (see Ref. [1]).

(b) **Constellation missions** with the objective of deploying multiple satellites to provide global service usually led by private organizations like companies or startups—Spire Inc. is a good example of the latter. Execution of these missions can be further divided into the following stages:

  i. **Technology demonstration stage** is the initial stage of a mission where a reduced number of satellites are deployed (usually a single satellite) to demonstrate the capacities of the proposed technology.

  ii. **Service provision stage** is the next stage of the mission, where the constellation is deployed through a set of "batches" or groups of satellites.

For single satellite missions a single ground station satisfies the basic requirements in terms of data downlink and command uplink for most of the use cases. In this sense, most institutions can execute their mission successfully with a single ground station, which can be upgraded or reused for future missions. However, most of the missions that can be included in this category are highly interested in getting more ground stations on the ground to increase the return of the mission.

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During the initial technology demonstration stage of constellation missions, a single ground station might also satisfy the necessities of these projects, in terms of amount data to be downlinked. When constellation missions get into the service provision stage, the necessity for developing a connected network of ground stations emerges. Satellite operators like Spire and Planet Labs have developed their own networks, with hardware and software specifically designed for their communications requirements. This way, their ground assets are very cost effective both in terms of development, deployment, and maintenance. The appearance of ground station network service providers has permitted outsourcing this task and eliminated the necessity of owning the ground assets.

1.2 Communication requirements

In terms of communication requirements, most CubeSat missions currently implement either one or more communication channels, for handling payload data separately from commanding and housekeeping/telemetry data. The reason for this is the very different requirements for the required data rate among these types of data. Taking into account the requirements for the communication systems for CubeSat missions as taken from Ref. [2], it is clear that satellite operators mostly choose the UHF band for their commanding and housekeeping/telemetry data and S band for their payload data. X band has become more popular over the last years, due to the emergence of new radio payload communication devices that were not available before. However, the utilization of X band is still reduced to downlinking data, and it is still prohibitive in terms of pointing requirements and power consumption for very small platforms—especially for 1U and 2U CubeSats. A good example of the utilization of X band in CubeSats is GomSpace satellite GOM-X3 (see Ref. [3]), which downlinked more than 115 MB of data in a 5.7-min pass. Planet Labs DOVE satellites have been repeatedly reported to be able to downlink data at a sustained rate of 200 Mbps (see Ref. [4]).

University projects mostly use amateur frequencies that are available for academic purposes (see Ref. [5]). This is a fast and affordable way for universities to get a frequency assigned for their satellite, but this approach does not allow operators to uplink telecommands through unattended remote ground stations. This restriction set by IARU limits the capacity of using remote ground stations for data collecting from satellites, when the data transfer protocols require a bidirectional satellite-to-ground communication.

1.2.1 Gain and pointing requirements

Pointing requirements depend on the selected band, that drives the selection of hardware for the ground station, especially when it comes to choosing the rotators and the mounting system for the antennas. For the VHF and UHF bands, cross Yagi antennas are the main choice for satellite operators, since they are the ones who work the best within this range of frequencies. In the case of Yagi antennas, the half power beamwidth ranges from 60 (VHF band) to 30 degrees (UHF band). In this sense, considering 3 dB of losses due to pointing guarantees the feasibility of the mission in terms of link budget, even with a drift in pointing of up to 15 degrees for the UHF band. For the UHF Yagi antennas, the gain usually ranges between 12 and 18 dBi (see Ref. [6] as an example of a UHF Yagi antenna). In the case of S band or X band, parabolic dishes are the main solution used at the ground stations. The gain and the half power beamwidth of a dish antenna follow the next well-known expressions:

$$g = e \cdot (\pi d/\lambda)^2 \tag{1}$$

$$\Theta = k\lambda/d = k/d \cdot c/f \tag{2}$$

where λ is the wavelength; d is the diameter of the dish; e is the aperture efficiency, ranging from 0.55 to 0.7; and k is a parameter that characterizes the performance of the antenna, which ranges between 70 and 50 depending on the manufacturing specifics for each antenna. The previous equations are plotted in Fig. 1, using the following reference values: frequencies 2.2, 8, and 27 GHz (S, X, and Ka bands); an aperture efficiency of $e = 0.5$; and a performance parameter with value $k = 70$. Analyzing these plots, we can conclude that the higher the required data rate is, the larger the diameter of the antenna for the same amount of transmitted power. As higher frequencies with higher bandwidth assignations are required to achieve higher data rates, the pointing requirements become more demanding, and the pointing systems become more complex. For S-band systems the pointing requirements can be less than 1 degree of precision, an achievable requirement using open-loop pointing systems for both

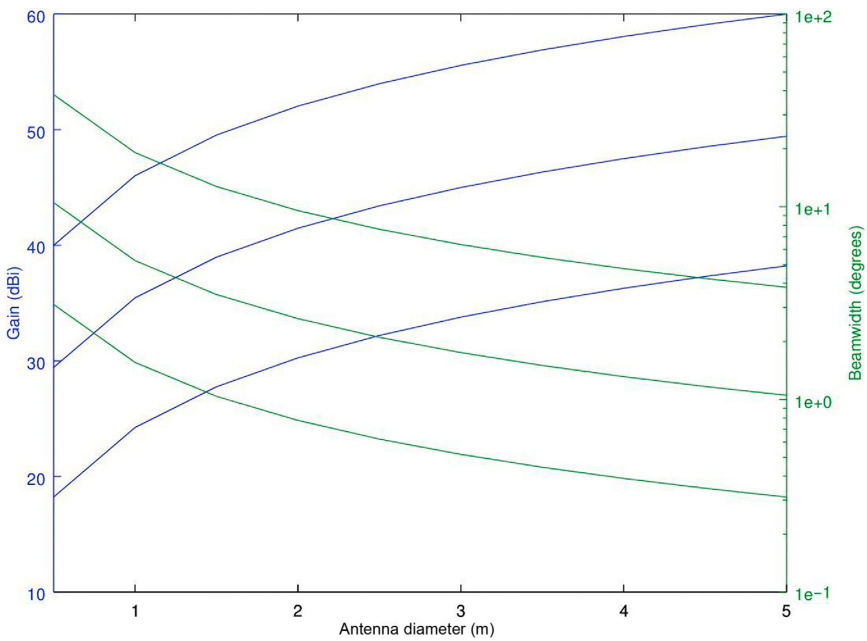

Fig. 1 Gain and beamwidth for S, X, and Ka band dishes.

communication ends. For Ka communication systems the beamwidth of the antenna goes well below 1 degree, potentially requiring closed-loop pointing systems (see Ref. [7]). X-band systems might require autotracking antennas or closed-loop systems for pointing or not depending on the mission.

1.2.2 Link availability and rain fade

When it comes to higher frequency links, the required system availability needs to be taken into account. It is necessary to include in the link budget the proper amount of attenuation due to rain fade, depending on how often the link with the satellite can be unavailable due to heavy rain. To properly account for this factor, it is recommended to follow the ITU recommendation (ITU P.838; see Ref. [8]). As per that recommendation, rain fade attenuation within the UHF and VHF bands is not a major concern, and its effects can be modeled by adding a fixed attenuation in the link budget calculations for any required system availability (see Ref. [9]). For X band and above the rain fade attenuation needs to be calculated, taking into account the location of the ground station itself and the required link availability. For those use cases in which neither the location nor the hardware cost can be increased, the availability of the link might need to be sacrificed.

1.2.3 Impact at system level

Taking into account these variables, a 3-m dish operating in S band will achieve around 30 dBi of gain and will require a pointing better than its beamwidth that should be slightly above 5 degrees. The same dish operating in X band will achieve around 45 dBi but requires an accuracy better than 0.5 degrees. It is easy to see that missions limited in terms of budget will easily opt for S-band-based ground assets rather than X-band.

2 Networked ground stations

Ground stations networks (GSNs) for larger satellites have existed for a longer time; however, their cost and performance are too high—and, therefore not appropriate for small satellite missions. These networks were also usually based on standards for ground station interconnection like SLE (Space Link Extension, see Ref. [10]). Ground station networks for small satellites started being developed, and they were mostly led by people coming from the IT industry. This context permitted the development of software-oriented infrastructures, based on software frameworks and tools coming from the IT industry.

The idea of networking ground stations for CubeSat missions was initially promoted by universities who were seeking to increase the scientific return of their missions. Several projects started in the United States (Mercury, GSN, and SatNet), together with the GSN project in Japan. The Education Office of the European Space Agency started a project called GENSO, which promoted the international cooperation in between different institutions from Europe, the United States, and Japan.

After this initial start as academic projects, several companies (Spire Inc. and Planet Labs) decided to develop their own networks of ground stations. The hardware and software that they use are well adapted to the specific requirements for their missions. Slightly later in time, ground station network providers like RBC Signals, Infostellar, and Leaf Space appeared in the market, proposing novel solutions based on the aggregation of existing resources (RBC Signals and Infostellar) or in the development of ground stations adapted to small satellite missions (Leaf Space). Companies like KSAT who offered products for larger satellites started developing their own solutions for small satellite missions like the KSAT lite network (see Ref. [11]).

2.1 University ground station networks

The Mercury (United States), GSN (Japan), GENSO (ESA led project), SatNet (United States and Spain) university-based ground networks are described below.

The Mercury and the GSN ground station networks were originally developed during the early 2000s in the United States and Japan, respectively. The Mercury network (Fig. 2) developed a client/server-based architecture and standardized the message exchange among ground stations through XML. The GSN network (Fig. 3) was composed of two services: the Ground Station Management Service (GMS) and the GS Remote Operation Web Service (GROWS). The GENSO project (Fig. 4) was a

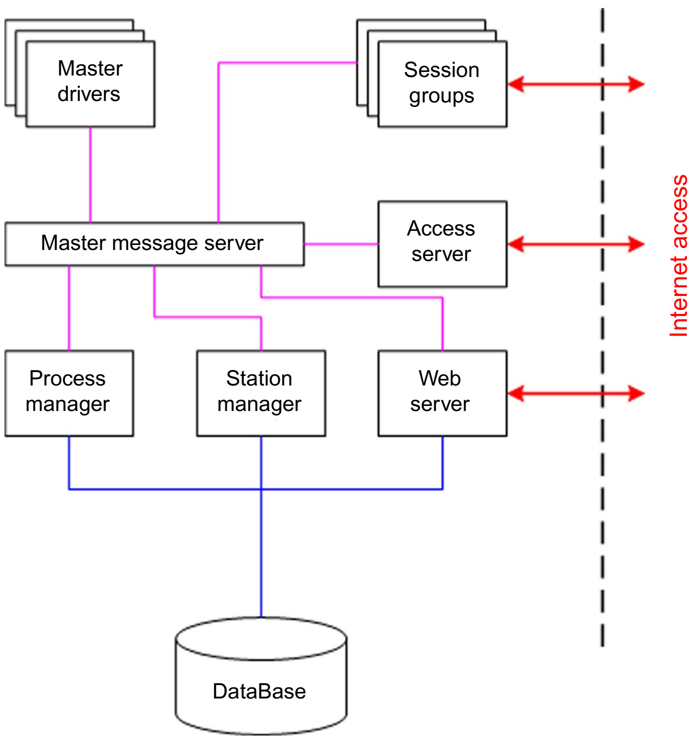

Fig. 2 Mercury GSN architecture.

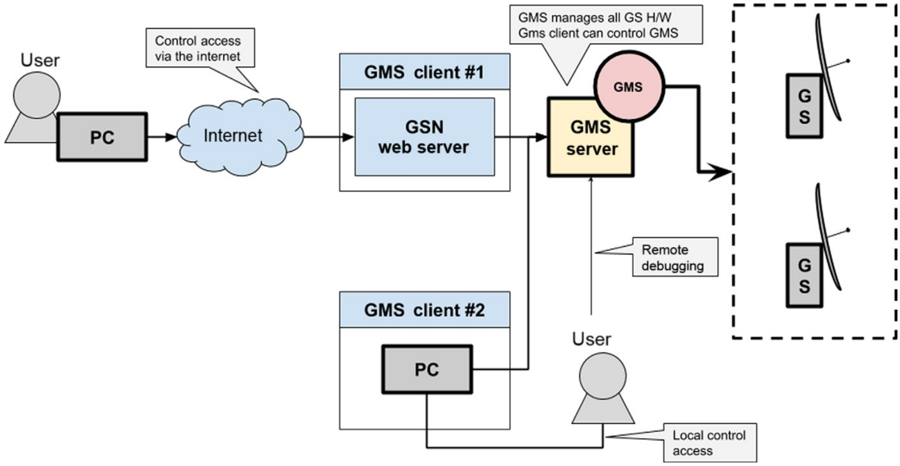

Fig. 3 GSN main architecture.

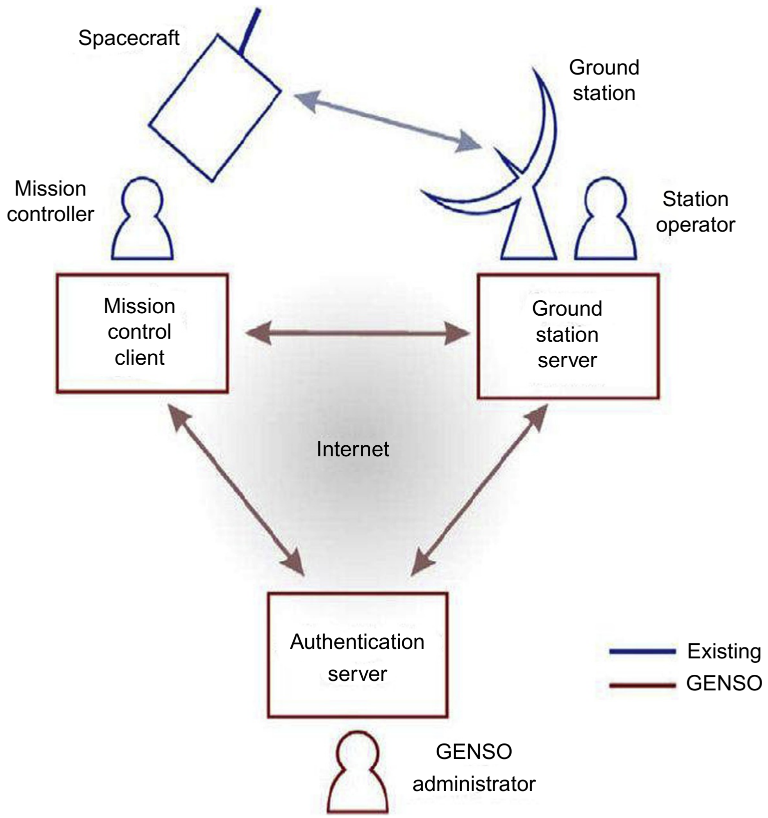

Fig. 4 GENSO architecture.

program of the Education Office of the European Space Agency, which connected several universities from Europe, Japan, and the United States. Its main objective was fostering the development of an educational ground station network to be used for supporting academic, scientific, and educational missions (see Ref. [12]). The implementation of the reference software was led by several universities, and it was based on a peer-to-peer (P2P) network architecture. The SatNet project (Fig. 5) started as a postdoctoral research fellowship granted by the Fundación Pedro Barrié de la Maza. The architecture of this network used IT industry software frameworks (Django, AngularJS, and TwistedMatrix) to develop a distributed server that could aggregate several remote ground stations using software integration only (see Ref. [13]).

Table 1 compares these networks in terms of high level features, from the point of view of the services offered to a satellite operator. In terms of network architecture paradigm, each architecture has been heavily influenced by the dominating network paradigm of its time.

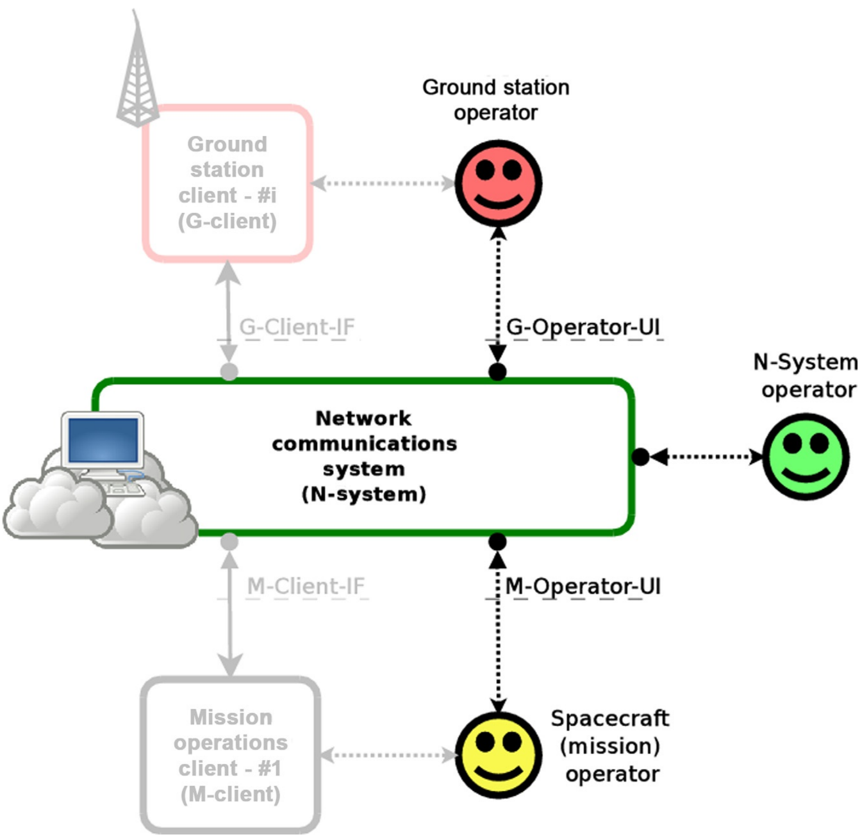

Fig. 5 SatNet architecture.

Table 1 Feature comparison.

Parameter/ GSN	Mercury	GSN	GENSO	SatNet
Paradigm	Client-server	Client-server	P2P	Distributed
Data transmission	Demodulated bitstream	Demodulated bitstream	Baseband RF in audio format	Demodulated bitstream
Scheduling	Direct usage	Direct usage	Distributed	Distributed
License	GPLv2	–	–	Apache v2

From R. Tubio, et al., Distributed operations network—first deployment results, in: 7th European CubeSat Symposium, doi:10.13140/RG.2.2.26165.99044.

2.2 Company ground station networks

The Planet Labs (United States) and Spire (United States) dedicated ground network stations are described below.

Planet Labs Inc. and Spire Global Inc. are two of the pioneering companies from the United States that started developing CubeSat-based constellations for either Earth observation (Planet) or for M2M/AIS/Space weather (Spire). When they started their missions (Planet started in 2010 and Spire started in 2012), there were no ground station network service providers that could meet their requirements in terms of cost and performance.

In this sense, Planet Labs Inc. developed its network to target the specific needs of its own satellites that operate in the UHF, S, and X bands (see Ref. [14]). This network is composed of 32 UHF Yagi stations, 11×5 m dishes and 7×7.6 m dishes, which allows Planet Labs to downlink around 700 GB of data per day, with a pass failure rate around 30%. In this sense the network of ground stations is deployed to increase the collection capacity from their satellites, which grew from a factor of 1.5 to a factor of 50 from 2015/Q3 to 2016/Q3 (see Ref. [15]).

Spire Inc. started deploying a network for their ground stations in San Francisco in 2012, expanding to more than 30 sites in a few years (see Ref. [16]). As of 2019, Spire is exploring the possibility of integrating their ground segment software with Amazon Web Service (AWS) ground station services. This will allow them to develop all their ground segment infrastructure in a serverless fashion, leaving the ground stations as the only physical assets that they have to manage outside the cloud infrastructure.

To sum up the modern approach for small satellite operators to develop the ground infrastructure is to rely on existing cloud technology. This technology permits satellite operators to write software that can be deployed serverless in the cloud and use all the existing resources of cloud providers like Amazon or Google to transfer and process data. In this sense the advantages of the cloud infrastructure are hard to match by any privately deployed system, especially in terms of cost, storage space, and computational power. On top of this, cloud systems like Amazon WS or Google Cloud provide ready-to-go frameworks that simplify the development and management of software applications for the cloud. At the end of the chain, API development for accessing the results of the processed data permits a fast interconnection with customer's systems (Figs. 6 and 7).

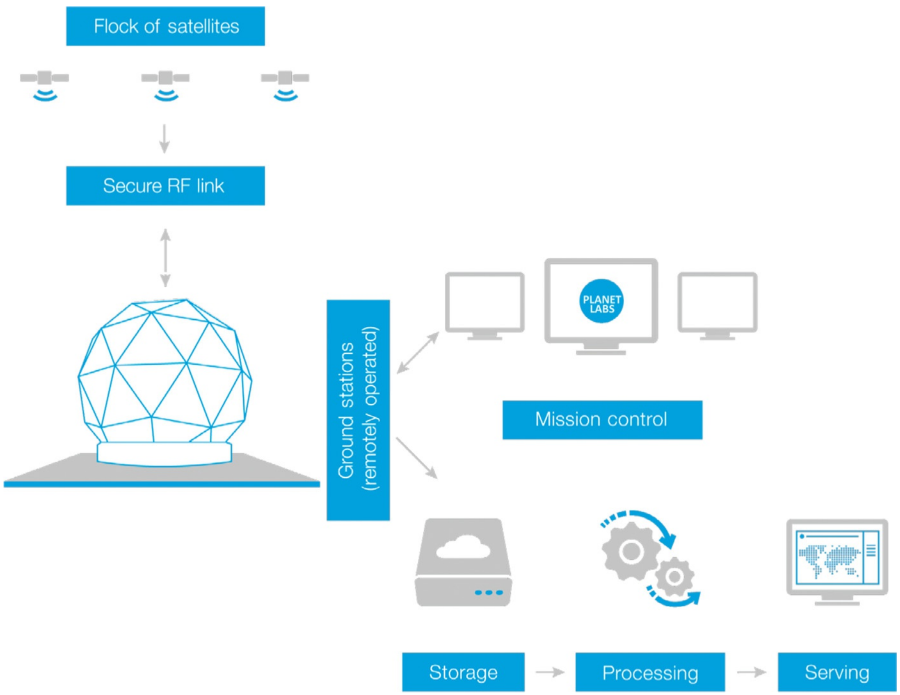

Fig. 6 Planet Labs ground segment architecture.
Courtesy of Planet Labs, data taken from Planet Labs Specifications: Spacecraft Operations & Ground Systems, version 1.0, June 2015. http://content.satimagingcorp.com.s3.amazonaws.com/media/pdf/Dove-PDF-Download.

2.3 Commercial ground station networks

The commercial ground station networks- Infostellar, RBC Signals, Leaf Space, KSAT, AWS Ground Stations, and SSC are described below.

Several companies started offering commercial ground station network services for the small satellite operators. These companies can be divided into two groups: companies who already were providing commercial services for larger satellites like KSAT (Kongsberg Satellite Services) or Swedish Space Corporation (SSC) and companies who were created to specifically target the necessities of smaller satellite operators, like Infostellar, RBC Signals, and Leaf Space. AWS Ground Stations started as part of the private space development program of Amazon. Existing ground station networks like KSAT or SSC started developing several products like KSAT lite and SSC Infinity. These products offer access through standard web API interfaces for the transmission of data and the execution of satellite operations. Both solutions standardize the amount of available options for the communications stacks and, at the

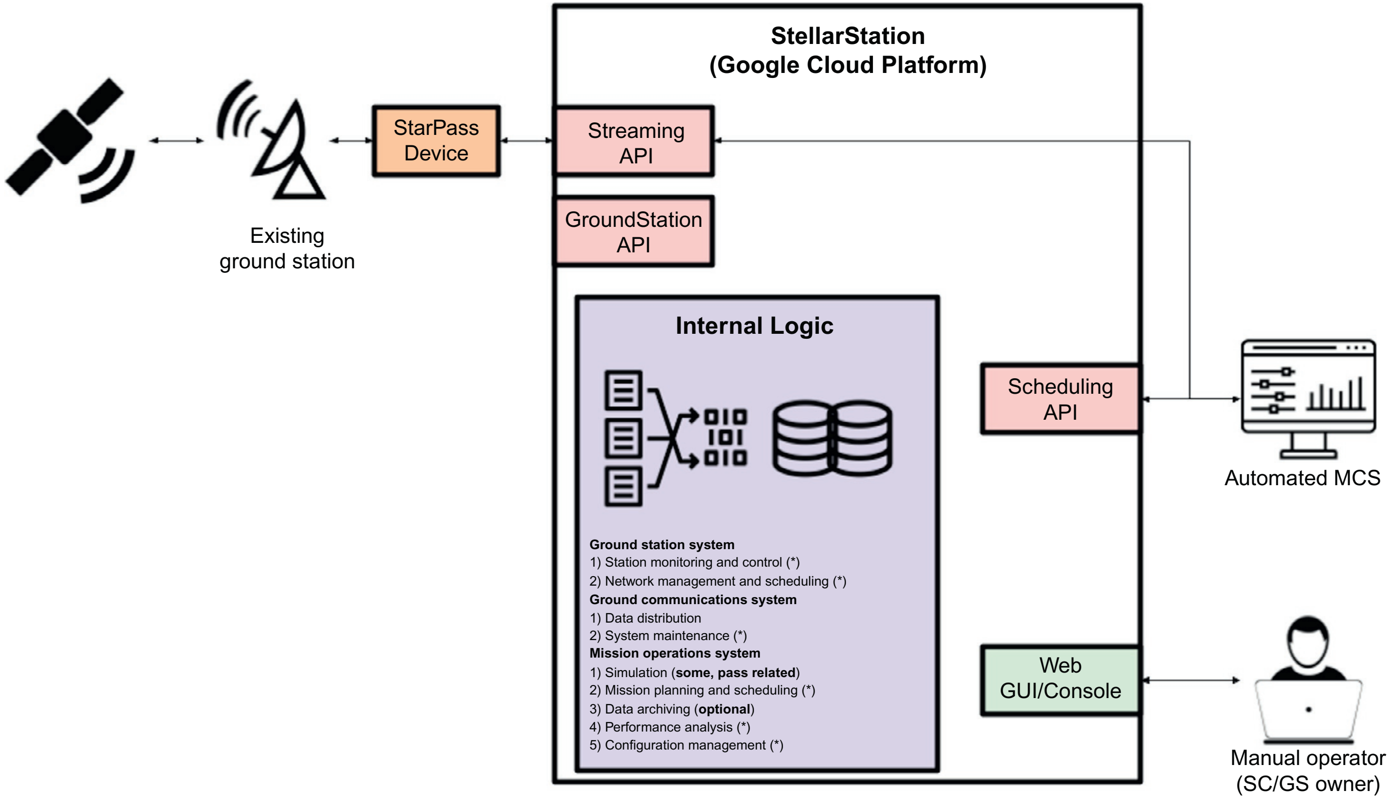

Fig. 7 Infostellar's ground station network architecture.

same time, look for providing standardized testing solutions to ease the satellite-to-ground integration process.

Other companies like Infostellar and RBC Signals started from the business perspective of aggregating existing ground stations and offering their unused capacity through the Internet APIs. Both companies do not develop or own any ground station asset as part of their core business; they only integrate existing ones. In this sense, both companies excel at the speed of integration of ground stations, and they usually look for becoming compatible with as many hardware manufacturers as possible. Leaf Space is a company that started both focusing on services for small satellite operators, and on developing their own ground stations network. In this sense the core of their business is split into creating specific networks for small satellite operators using their own design for the ground stations and offering access to their proprietary-shared low-cost network for fast seamless integration.

These new companies grew their networks using existing cloud infrastructure as the core of their systems. This allows them to integrate third parties API-based mission operation system (MOS) very easily, leveraging the complexity of developing the ground segment for a space mission. Implementing the GSN and MOS bundled within the cloud permits satellite operators to focus on the processing of the payload data and on the development of data-based services. With this approach, satellites and ground stations become commodities, and the focus shifts toward the development of big data and machine learning applications.

3 Conclusions

During the last several years, ground station networks have evolved toward cloud-based software systems that come bundled with mission operation systems. Hardware centric-based ground station networks have evolved toward systems that rely on the existing commercial infrastructure for service provision. The integration process relies on the usage of software-defined radios and automated software-based compatibility tests. At the same time, instead of focusing on providing solutions that meet all the specific requirements of the customer, these networks provide standardized communications stacks that ease the compatibility process. Space ground infrastructure is moving toward a service-based business, to which satellites get connected to ground stations like mobile phones seamlessly connect to our current cell telephony networks.

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